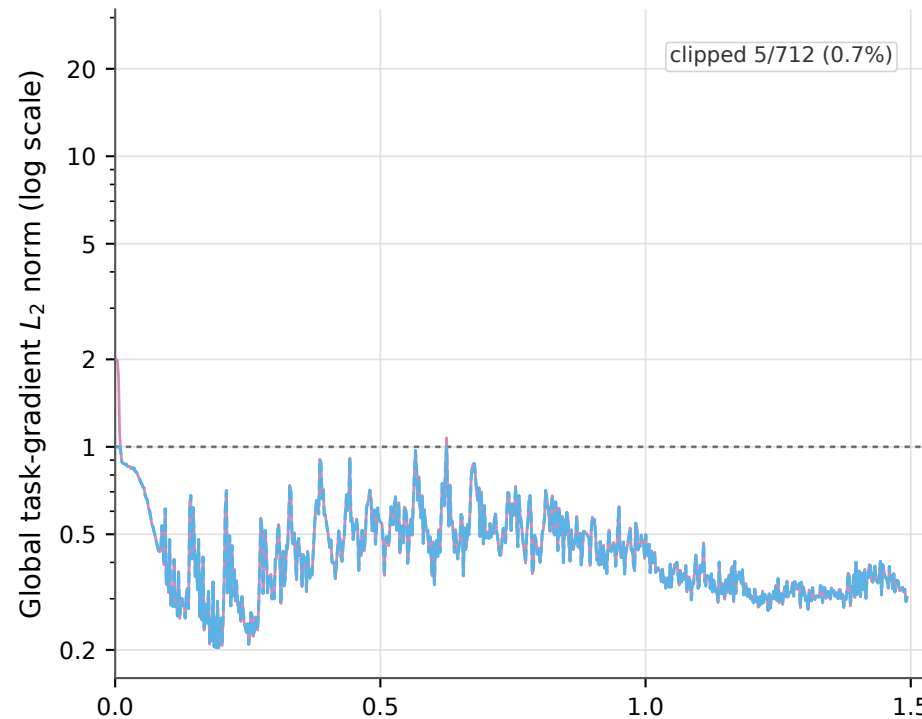


A0 task-gradient norms through one MiniPile pass

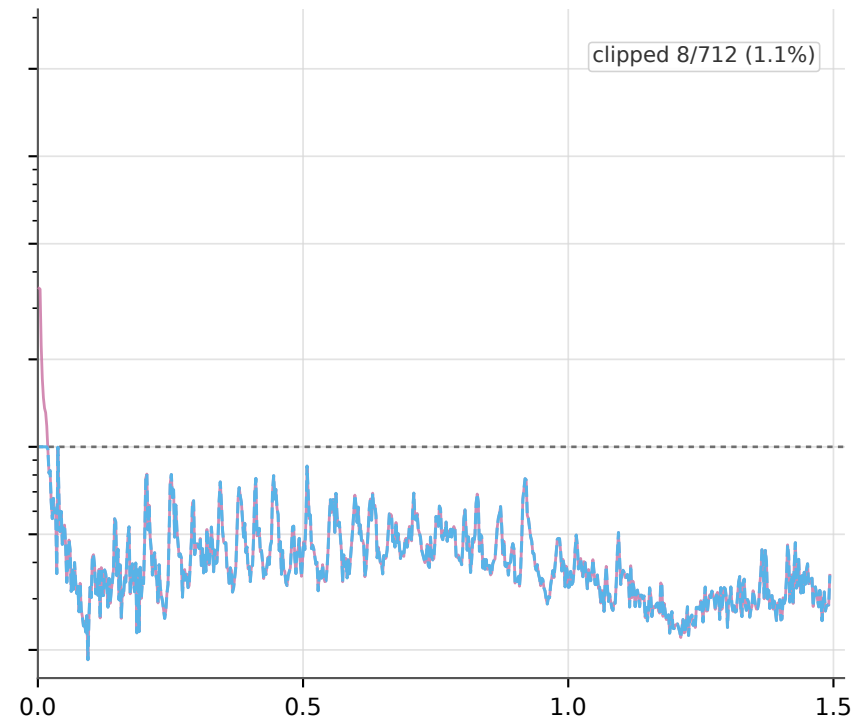
Global full-model L2 norm at every optimizer boundary; shared axes across model sizes

— Before clipping - - - After clipping - - - Global clip threshold = 1

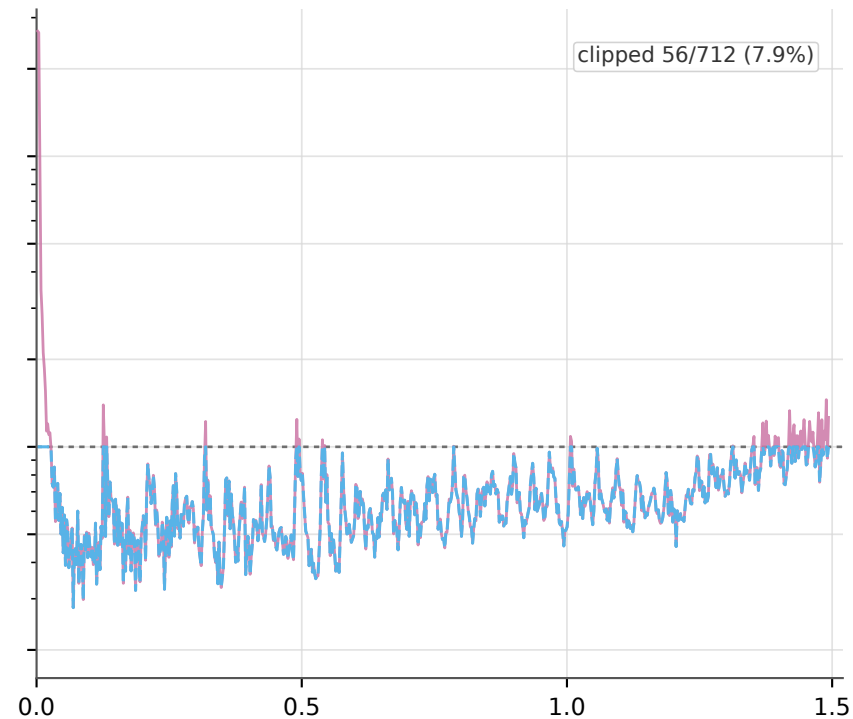
Pythia-14M



Pythia-70M



Pythia-410M



Raw boundary values; no smoothing. Each boundary contains 2,097,152 tokens. All 712 updates completed without overflow or skipping.
The norm is not parameter-count normalized; identical global clipping at 1.0 therefore need not have identical effects across scales.